

Assignment 3 — Finite Difference Schemes for the Linear Advection Equation

Numerical Methods for Partial Differential Equations

Released: 27 March 2026 **Due:** 02 April 2026 **Total marks:** 50

Language: C/C++ for code and Python for images Submit: one code file *per problem* + one PDF report

You will implement seven classical finite difference schemes for the 1-D linear advection equation. Check the boxes below for the problem you will be solving. Each student is assigned a unique Courant number ν via their roll number.

General instructions

- Read your Courant number ν from the roll-number table inside each problem. Do *not* change it, the variability is intentional. Wrong choice as per your roll number will mean zero marks.
- Use the **common parameter set** below for all five problems unless stated otherwise. Wrong choice as per your roll number will mean zero marks.
- Plot your numerical solution alongside the exact solution at the specified output times.
- If your scheme appears to blow up, still submit the plot (up to the last stable time step or after 200 steps if it diverges immediately). Comment on what you observe.
- **Save every script.** You will reuse and extend them later.

Common parameter set for all problems: 1-12 roll numbers

PDE:	$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0$	Domain:	$x \in [0, 1]$, periodic BCs
Speed:	$c = 1 \text{ m s}^{-1}$	Grid:	$N_x = 50$, $\Delta x = 1/50 \text{ m}$
Init. cond.:	$u(x, 0) = \exp(-100(x - 0.25)^2)$	Final time:	$T = 0.5 \text{ s}$
Exact soln:	$u_{\text{ex}}(x, t) = \exp(-100((x - t) \bmod 1 - 0.25)^2)$	Time step:	$\Delta t = \nu \Delta x / c$
Output times:	$t = 0, 0.125, 0.25, 0.375, 0.5 \text{ s}$. L_2 error: $\ e\ _2 = \sqrt{\Delta x \sum_i (u_i^n - u_{\text{ex},i})^2}$ at $t = T$.		

Common parameter set for all problems: 13-25 roll numbers

PDE:	$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0$	Domain:	$x \in [0, 1]$, periodic BCs
Speed:	$c = 1 \text{ m s}^{-1}$	Grid:	$N_x = 100$, $\Delta x = 1/100 \text{ m}$
Init. cond.:	$u(x, 0) = \sin\left(\frac{2\pi x}{L}\right)$	Final time:	$T = 1 \text{ s}$
Exact soln:	$u_{\text{ex}}(x, t) = \sin(2\pi(x - ct)/L)$	Time step:	$\Delta t = \nu \Delta x / c$
Output times:	$t = 0, 0.25, 0.5, 0.75, 1 \text{ s}$. L_2 error: $\ e\ _2 = \sqrt{\Delta x \sum_i (u_i^n - u_{\text{ex},i})^2}$ at $t = T$.		

Problem 1. FTBS and FTFS — First-Order Upwind Schemes**[10 marks]**

This problem implements and compares two first-order explicit one-sided schemes for the advection equation with $c > 0$.

Schemes.

Forward-Time Backward-Space (FTBS / upwind):

$$u_i^{n+1} = u_i^n - \nu(u_i^n - u_{i-1}^n) \quad (1)$$

Forward-Time Forward-Space (FTFS / downwind):

$$u_i^{n+1} = u_i^n - \nu(u_{i+1}^n - u_i^n) \quad (2)$$

Your Courant number ν is given by the value in the table below.

Roll	1	2	3	4	5	6	7	8	9	10	11	12	13
ν	0.30	1.74	0.42	1.62	0.54	1.50	0.66	1.38	0.78	1.26	0.90	1.14	1.02
Roll	14	15	16	17	18	19	20	21	22	23	24	25	
ν	1.08	0.96	1.20	0.84	1.32	0.72	1.44	0.60	1.56	0.48	1.68	0.36	

Tasks.

- Implement *both* FTBS and FTFS using the common parameter set. If periodic boundary conditions: index $i - 1$ wraps to $N_x - 1$ and $i + 1$ wraps to 0 when needed. [5]
- Plot $u(x, t)$ at the five output times for both schemes on the same set of axes (one figure per scheme), overlaid with the exact solution. [2]
- Compute and report $\|e\|_2$ at $t = T$ for FTBS. [2]
- In one short paragraph: what physically causes FTFS to behave the way it does for $c > 0$? Compare the numerical diffusion of the two schemes. [1]

Problem 2. FTCS — Centred-Space Scheme**[10 marks]**

The Forward-Time Centred-Space scheme replaces the spatial derivative with a symmetric second-order finite difference:

$$u_i^{n+1} = u_i^n - \frac{\nu}{2}(u_{i+1}^n - u_{i-1}^n) \quad (3)$$

Your Courant number:

Roll	1	2	3	4	5	6	7	8	9	10	11	12	13
ν	0.25	1.40	0.35	1.30	0.45	1.20	0.55	1.10	0.65	1.00	0.75	0.90	0.85
Roll	14	15	16	17	18	19	20	21	22	23	24	25	
ν	0.80	0.95	0.70	1.05	0.60	1.15	0.50	1.25	0.40	1.35	0.30	1.45	

Tasks.

- Implement FTCS with the common parameter set and your assigned ν . [3]
- Run the simulation. Plot $u(x, t)$ at the five output times. If the solution diverges, plot the first 50 time steps, clipping the y -axis to $[-5, 5]$ so the oscillation growth is visible. [3]

- (c) Compare the behaviour you observe for small ν (e.g. rolls 1–5) versus large ν (e.g. rolls 20–25). Does a smaller Courant number rescue the scheme? [3]
- (d) Compute the growth rate of the maximum absolute value $\|u\|_\infty$ over the first 200 time steps and plot it on a log scale. [1]

Problem 3. Lax–Friedrich Scheme**[10 marks]**

The Lax–Friedrich scheme replaces the centred time level u_i^n with the spatial average of its two neighbours:

$$u_i^{n+1} = \frac{1}{2}(u_{i+1}^n + u_{i-1}^n) - \frac{\nu}{2}(u_{i+1}^n - u_{i-1}^n) \quad (4)$$

Your Courant number:

Roll	1	2	3	4	5	6	7	8	9	10	11	12	13
ν	1.60	0.45	1.50	0.55	1.40	0.65	1.30	0.75	1.20	0.85	1.10	0.95	1.00
Roll	14	15	16	17	18	19	20	21	22	23	24	25	
ν	1.05	0.90	1.15	0.80	1.25	0.70	1.35	0.60	1.45	0.50	1.55	0.40	

Tasks.

- (a) Implement Lax–Friedrich with the common parameter set. [3]
- (b) Plot the solution at the five output times alongside the exact solution. Describe the dominant error character: is it diffusive or dispersive? [3]
- (c) Compute $\|e\|_2$ at $t = T$. Compare with the FTBS error (Problem 1) at the same ν if possible, or at $\nu = 0.5$ for both. Which scheme is more accurate and why? [3]
- (d) Plot $\|e\|_2$ as a function of Δx for $N_x \in \{25, 50, 100, 200\}$ (keeping ν fixed) on a log-log scale. Determine the observed convergence rate. [1]

Problem 4. Lax–Wendroff Scheme**[10 marks]**

The Lax–Wendroff scheme achieves second-order accuracy in both space and time by including a corrective term derived from the Taylor expansion:

$$u_i^{n+1} = u_i^n - \frac{\nu}{2}(u_{i+1}^n - u_{i-1}^n) + \frac{\nu^2}{2}(u_{i+1}^n - 2u_i^n + u_{i-1}^n) \quad (5)$$

Your Courant number:

Roll	1	2	3	4	5	6	7	8	9	10	11	12	13
ν	0.50	0.56	0.62	0.68	0.74	0.80	0.86	0.92	0.98	1.04	1.10	1.16	1.22
Roll	14	15	16	17	18	19	20	21	22	23	24	25	
ν	1.28	1.34	1.40	1.46	1.52	1.58	1.64	1.70	1.76	1.82	1.88	1.94	

Tasks.

- (a) Implement Lax–Wendroff and run to $t = T$. [5]
- (b) Plot the solution at the five output times. You should observe Gibbs-like oscillations trailing the pulse for stable runs. Explain their physical origin in terms of the scheme’s dispersion relation. [3]

- (c) Compute $\|e\|_2$ at $t = T$ and compare it with the Lax–Friedrich result (Problem 3) at the same ν . Tabulate the comparison and explain the difference in accuracy. [2]

Problem 5. MacCormack and Crank–Nicolson

[10 marks]

This problem implements a predictor–corrector explicit scheme and an unconditionally stable implicit scheme, and compares their behaviour at the same ν .

MacCormack scheme (explicit, predictor–corrector):

$$\tilde{u}_i = u_i^n - \nu(u_{i+1}^n - u_i^n) \quad (\text{forward-space predictor}) \quad (6)$$

$$u_i^{n+1} = \frac{1}{2}[u_i^n + \tilde{u}_i - \nu(\tilde{u}_i - \tilde{u}_{i-1})] \quad (\text{backward-space corrector}) \quad (7)$$

Crank–Nicolson scheme (implicit, time-centred):

$$u_i^{n+1} + \frac{\nu}{4}(u_{i+1}^{n+1} - u_{i-1}^{n+1}) = u_i^n - \frac{\nu}{4}(u_{i+1}^n - u_{i-1}^n) \quad (8)$$

Equation (8) forms a tridiagonal (circulant for periodic BCs) linear system $\mathbf{A} \mathbf{u}^{n+1} = \mathbf{b}^n$ that must be solved at each time step.

Your Courant number:

Roll	1	2	3	4	5	6	7	8	9	10	11	12	13
ν	0.60	0.67	0.74	0.81	0.88	0.95	1.02	1.09	1.16	1.23	1.30	1.37	1.44
Roll	14	15	16	17	18	19	20	21	22	23	24	25	
ν	1.51	1.58	1.65	1.72	1.79	1.86	1.93	2.00	2.07	2.14	2.21	2.28	

Tasks.

- (a) Implement *both* schemes. [6]
- (b) Run both schemes to $t = T$ with your assigned ν . Plot the solutions at the five output times on a single figure (two subplots, one per scheme), including the exact solution for comparison. [2]
- (c) Tabulate $\|e\|_2$ at $t = T$ for both schemes. For students with $\nu > 1$: MacCormack will be unstable — report the Crank–Nicolson error only and note the MacCormack failure. [2]
- (d) For Crank–Nicolson only: increase ν to $\{2, 5, 10\}$ (keeping Δx fixed, so Δt grows large). Plot the solutions and comment on accuracy versus stability. Does unconditional stability mean unconditional accuracy? [2]

Appendix B — Summary

Complete this table below.

Scheme	Order $(\Delta t, \Delta x)$	Stability condition	Character of error	My ν stable?
FTBS				
FTFS				
FTCS				
Lax–Friedrich				
Lax–Wendroff				
MacCormack				
Crank–Nicolson				